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ReasoningV: Efficient Verilog Code Generation with Adaptive Hybrid Reasoning

Haiyan Qin, Jingjing Li, Zhiwei Xie, LiangChen Li, Xiaotong Feng, Junzhan Liu, Wang Kang

ABSTRACT

Large Language Models (LLMs) have advanced Verilog code generation significantly, yet face challenges in data quality, reasoning capabilities, and computational efficiency. This paper presents **ReasoningV**, a novel model employing a hybrid reasoning strategy that integrates trained intrinsic capabilities with dynamic inference adaptation for Verilog code generation. Our framework introduces three complementary innovations: (1) ReasoningV-5K, a high-quality dataset of 5,334 functionally verified instances with reasoning paths from PyraNet filtering; (2) a two-stage training approach combining LoRA for foundational knowledge with full-parameter optimization for reasoning; and (3) an adaptive reasoning mechanism that dynamically adjusts reasoning depth based on problem complexity, reducing token consumption by **78%** while preserving performance. ReasoningV offers a more reliable and accessible pathway for advancing AI-driven hardware design automation, with our model, data, and code available at *https://github.com/BUAA-CLab/ReasoningV*.

HIGH-PERFORMANCE VERILOG CODE GENERATION FRAMEWORK

Innovation 1: High-Quality Reasoning Dataset Construction

Innovation 2: Common Sense and Reasoning Enhancement Training

Innovation 3: Adaptive Reasoning (359% Efficiency

METHODOLOGY

Multi-stage Data Filtering Pipeline

Step 1: Multi-Dimensional Filtering

Step 2: Reasoning Enhancement

Step 3: Final Dataset Assembly

Finally, from 690K PyraNet items to 5,334 verified samples. We cut 99.3% noise yet retain diversity across 18 categories.

Code-Free Reasoning Paths

We adopted a "problem-description-reasoning path-solution-testbench" format. Each entry includes the problem description, a detailed reasoning path, the Verilog solution, and a corresponding functional testbench. Finally,functional verification was performed by simulating each solution against its testbench using Icarus Verilog. Samples failing simulation (approx. 23.3% of the 7K candidates) were discarded.

TWO-STAGE TRAINING METHOD

ReasoningV employs a two-stage training strategy: Stage 1 establishes foundational knowledge using parameter-efficient LoRA (1.05% of parameters), while Stage 2 develops deep reasoning via full-parameter optimization on ReasoningV-5K.

ADAPTIVE REASONING METHOD

A lightweight adapter categorizes each input into one of three difficulty levels: "Easy", "Medium", or "Hard" with token budgets 512/1280/4096, enabling fine-grained control over downstream reasoning strategies.

EXPERIMENTS

MAIN RESULTS

Performance comparison across Verilog code generation benchmarks.

Category	Model	VEval-H			VEval-M			RTLLM	
		P@1	P@5	P@10	P@1	P@5	P@10	P@1	P@5
Commercial	GPT-4o-mini	44.2	49.4	56.4	60.1	62.2	65.0	49.2	61.5
	DeepSeek-R1	**81.7**	**88.1**	**89.7**	**81.1**	85.8	**87.4**	**58.5**	**66.8**
	DeepSeek-V3	70.7	77.4	78.8	77.6	**86.2**	**87.4**	54.9	63.7
	Gemini-2.0	59.5	67.8	70.9	74.9	79.5	81.5	46.5	56.4
Open Source	Qwen-7B	34.3	46.3	50.0	61.1	75.7	78.6	31.3	45.2
	Qwen-14B	**48.2**	**59.8**	**63.5**	**64.9**	**77.1**	**79.7**	**41.6**	**54.1**
	Qwen-32B	47.6	59.0	61.5	61.5	64.1	75.5	40.2	**55.7**
Verilog	RTLCoder-DS	41.6	50.1	53.4	61.2	76.5	81.8	33.5	43.9
	BetterV-CQ	46.1	53.7	58.2	68.1	79.4	84.5	–	–
	CraftRTL-15B	**68.0**	**72.4**	–	**81.9**	**86.9**	–	49.0	65.8
	OriGen-DS	54.4	60.1	**64.2**	74.1	82.4	**85.7**	–	**65.5**
	OriGen-LoRA	47.4	59.9	63.4	70.2	80.2	82.7	**44.1**	54.0
RV (Ours)	RV-7B	57.8	69.3	72.4	73.6	83.4	85.3	44.6	62.2
	RV-14B	**73.9**	**83.7**	**85.9**	**75.6**	**85.2**	**87.4**	**50.6**	**64.7**

Note: VEval-H = VerilogEval-human, VEval-M = VerilogEval-machine, P@k = pass@k, DS = DeepSeek, CQ = CodeQwen, – = not reported

RV-14B hits 73.9% pass@1 on VerilogEval-human;
RV-7B reaches 57.8% with strong efficiency.
On Verilog Eval-machine: 75.6% (14B). RTLLM: 50.6% (14B).
We narrow the gap to larger models and lead among accessible open-source contenders.

ABLATION STUDIES

Ablation study on the effectiveness of two-stage training methodology.

MODEL	VerilogEval-human	VerilogEval-machine	RTLLM
Base	34.3	61.1	31.3
RV-CT	51.3	64.5	42.5
RV-RET	45.5	65.7	33.3
RV-7B	**57.8**	**73.6**	**44.6**

Note: **CT** = Commonsense Training, **RET** = Reasoning Enhancement Training

Two-Stage works: Stage1-only (CT) < Stage1+2; Stage2-only (RET) can't compensate for weak domain grounding. Adaptive decoding matches Hard on machine benchmark with 75% fewer tokens. Takeaway: grounding + reasoning + routing wins.

Performance and efficiency of reasoning modes on Verilog benchmarks.

MODE	VerilogEval-H		VerilogEval-M		RTLLM	
	P@1	Tokens	P@1	Tokens	P@1	Tokens
DS-R1	81.7	6070 (+584%)	81.1	3311 (+680%)	58.5	12686 (+1377%)
Easy	32.1	**235** (-74%)	51.3	**164** (-61%)	29.7	**356** (-59%)
Medium	54.3	1302 (+47%)	68.4	1156 (+173%)	42.8	1650 (+92%)
Hard	**57.8**	2473 (+178%)	**73.6**	1725 (+307%)	44.6	3940 (+359%)
Adaptive(7B)	53.0	888	**73.6**	424	**45.4**	859

Note: **H** = human, **M** = machine, **P@1** denotes pass@1 metric, **DS** = DeepSeek

We allocate effortby difficulty, avoiding overthinking easy tasks.
Results: 85–93% token savings vs commercial reasoning models and 32–75% vs fixed-depth while preserving accuracy